

Task: POD - Travels

Lab 9. Available memory: 512 MB.

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Description

Bajtocja is a country with n cities. Unfortunately, building roads in Bajtocja is very expensive, so only $n - 1$ roads have been built to connect them. Each road is bidirectional, and the road network is such that it is possible to travel between any pair of cities, possibly passing through other cities along the way.

Recently, q inhabitants of Bajtocja approached the Bajtockie Travel Agency with inquiries. Each of them lives in a certain city but plans a vacation soon, during which they will visit two other cities. To do this, they will likely have to visit certain other cities along the way.

The inhabitants of Bajtocja really like fridge magnets, and for every city they find themselves in, they will buy one such magnet as a souvenir. However, if they visit a city again, they do not feel the need to buy a second one, nor do they feel the need to buy souvenir magnets from the city where they live.

Since the more magnets they buy, the larger a new fridge they will need to buy, they are interested in the total number of cities they will visit during their vacation. As an employee of the Bajtockie Travel Agency, you must help the inhabitants buy appropriately sized fridges by answering these questions.

Input

The first line of input contains two integers n and q ($1 \leq n \leq 200\,000$, $1 \leq q \leq 30\,000$), denoting the number of cities in Bajtocja and the number of inhabitants we must help.

In the following $n - 1$ lines, the description of the road network of Bajtocja is given, where the i -th of these lines contains two integers u_i, v_i ($1 \leq u_i, v_i \leq n$, $u_i \neq v_i$), meaning that there is a road between cities numbered u_i and v_i .

The next q lines contain the description of queries from inhabitants, where the i -th of them contains three distinct integers a_i, b_i, c_i ($1 \leq a_i, b_i, c_i \leq n$), meaning that the i -th inhabitant lives in city a_i and plans a trip from there to city b_i , from there to city c_i , and then back to city a_i .

Output

The program should print q lines. In the i -th line, print one integer denoting the number of different cities the i -th inhabitant will visit during their trip (excluding their city of residence).

Example

Input:

6 3
1 4
1 6
2 4
4 5
1 6
1 2 3
6 2 5
2 4 1

Output:

4
4
2